

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (ISE) (Credit System) Degree Examinations

May - June 2023

20ISE74 – COMPUTER VISION

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** and **Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Unit – I

1. a) List & describe any eight applications of computer vision with example images.
- b) What is image morphing and image stitching? Explain.
- c) What is motion capture and 3D modelling as in Bing maps? Explain.

Marks	BT*	CO*	PO*
8	L*2	1	1
6	L2	1	1
6	L2	1	1

2. a) Write the Geometric 2D transformations matrices for translation, rotation, scaling, flip along x-axis followed by flip along y axis and shear along x axis. And show it for a line joining P1-P2 P1(100,50) P2(200,200). $T_x=20$ $T_y=15$ $\Theta=90$ degrees(clockwise), $s_x=0.5$ $s_y=0.5$, shear along x-axis $sh_x=1.5$ and flip along x axis

10	L2	1	2
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- b) List the point operations on an image. Write the corresponding formula and demonstrate them with help of pictures.

10	L2	1	1
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3. a) Apply G 3x3 matrix on the image I (4x4) and calculate the resultant image I'(4x4)

1	0	-1
1	0	-1
1	0	-1

3X3

2	2	2	3
2	1	3	3
2	2	1	2
1	3	2	2

4X4

8	L2	2	2
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- b) Draw and explain how you will smooth and differentiate the following function.

$f(x)$

Fig 3(a)

4	L2	2	2
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- c) Write the steps in canny edge detection and briefly explain with sample pictures.

8	L2	2	1
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Unit – II

4. a) What is pinhole camera? Illustrate the mathematical model of a pinhole camera with neat diagrams and equations
- b) Explain epi-polar geometry with a neat diagram. Clearly mention epipoles, epipolar lines, epipolar plane and epi-polar pencil in it.

8	L2	3	1
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8	L2	3	1
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P.T.O.

c)	What is image rectification and mention any two of its applications.	4	L2	3	1
5. a)	Derive the camera matrix K of a digital camera.	6	L2	3	1
b)	Distinguish between intrinsic and extrinsic parameters of a digital camera.	6	L2	3	1
c)	Discuss the segmentation technique -mean shift clustering with an example and a diagram.	8	L2	4	1
6. a)	What is optical flow? Derive the Constraint equation of optical flow.	4	L2	4	1
b)	Explain lucas-kanade method to determine optical flow. Derive the structure tensor.	8	L2	4	1
c)	Describe an object tracking method involving SIFT features, search window, and match score with neat diagrams.	8	L2	4	1

Unit – III

7. a)	What is binary image matting? Explain with a sample image.	6	L2	5	1
b)	Explain Image parsing technique with an example.	6	L2	5	1
c)	Explain face recognition and its pipeline.	8	L2	5	1
8. a)	Write short notes on nearest neighbour match.	10	L3	5	1
b)	Write short notes on principal component analysis(PCA) and eigen faces.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome .
